ARTIFICIAL INTELLIGENCE

It is the branch of computer science, that deals with creating machine that are capable of act, think, make decisions, reasoning like humans.

Applications of AI

- Health care: analyze medical images (x-ray, MRI)
- Education: AI tutoring, online learing recommendation.
- Banking and finance: fraud detection, online Banking chatbots, credit scoring, and loan approval.
- Transportation: self-driving cars, navigation, traffic prediction.
- E-commerce: product recommendations, customer service chatbots.
- Agriculture: detect plant diseases, smart irrigation system, crop ~~manito~~ monitor
- Manufactering: Quality inspection, industrial robots.
- Entertainment: video and music recommendations, video game AI
- Security: face recognisation, finger print and voice authentication, cyber security, threat detection.
- Virtual assistants: Siri, BixBy, alexa, google assistance

Advantages:-

-> Reduces Human error
-> works 24/7
-> Increases efficiency - In repeatative tasks there by saving the time
-> Improve decision making
-> performs dangerous task
-> Enhances Health car
-> Better customer service.

Disadvantages

-> Job displacement
-> High development- cost
-> Lacks humans emotion
-> privacy concerns
->security risk - deep fake, cyber attacks etc
-> Bias and errors

Foundations of AI

1) Computer science
2) mathematics
3) philosophy (Ethics)
4) pschychology (studies how human thinks, learns and remembers)
5) Neuro science - study of brain
6) Linguistics - study of human language
7) Economics - study of maximising the benefits and minimising the cost
8) Robotics

History of AI

⇒ Founder - John Mc carthy in the year 1956.
⇒ 1950 - Turing test (Alan turing build it)
If a person cannot distinguish between a heman and machine through conversation the machine is considered as intellegent.
⇒ 1956 -> official beginning of AI as a field of study.
⇒ 1960 - 70 -> researchers develop programs that could solve puzzles and play simple games.
⇒ 1966 -> Eliza -> It is the first chatbots build
A AI chess is build in this year
⇒ 1974-1980 - first AI winter - AI research funding declined due to limited process

1980 -> AI was started to used in medicine, engineering and business
1987 - 1993 - second AI winter - AI funding and interset decreased again beause export systems were expensive and limited.
1997 -> Deep blue -> IBM deep blue defeated world chess champion harry kasparov.
2002 -> Roomba -> first successful AI powered AI robotic created.
2011 -> IBM watson -> watson won the quiz by complex questions.
2016 -> Alphago -> deep minds Alphago defeated world go champion lee sedol.
2022 -> chat GPT.

Agent

Agent in artificial intelligence is an entity that perceives its environment through sensor analyses it and takes action upon the environment through actuators to achive specific goal.
Ex:- A self driving car is a AI agent
sensors :- camera, GPS, redor
Agent program- Analyse road conditions and traffic
Actuators - Break, staring,

Structure of agent.
Structure of AI agent describes how an agent is organised to observe the environment process the information, make decision and perform actions to achieve a goal.
-It has two main components
1) agent program (software or brain)
2) Architecture (agent machine / Hardware)
Agent = Architecture + agent program.

structure diagram.

Environment
↓
sensor
↓
agent program (software)
↓
Runs on architecture
↓
Actuator
↓
Environment

Environment: Everything around the agent with which it interacts .. It provides information to the agent & it is affected by agents action

Ex:- Chessboard for a chess playing AI, House for a robot vaccum cleaner, roads for self driving car.

Sensor: sensors are the device that collect information from the environment.
They allow AI agent to know what is happening around it,

Ex:- Camera, GPS, radar, temperature sensors.

Functions

- collect data from the environment
- send information to the agent program.

Agent program (brain or software) -
It is the software that recieves information from the sensor process it makes decision.
It applies AI Algorithms or rules to take the decision.

Architecture (machine / hardware)
It is the hardware platform on which agent program runs.

Actuator (output device)
Actuators are the devices that carry out the actions decided by agent program.
ex:- motor, Robot arm, speaker, display device.

workflow of self driving car.

1. Suppose camera detects a red traffic signal
2. The information is send to agent program
3. The agent program decides to stop
4. The architecture executes the decision
5. The breaks are applied
6. The car stops safely.

Task environment

-> Is the situation in which AI agent works to achieve a specific goal.
-> It describes what agent must do where it works how its performance is measured and how it interacts with its surrounding.

PEAS structure.

P -> performance
E -> environment
A -> Actuator
S -> sensor.

performance - safety, comfort, milage.

Vcum cleaner -> P -> cleaning or clean all dirt, use less electricity, finish quickly.
E -> floor, furniture, dust, room, wall
A -> wheels, brushes, vaccum motor
S -> camera, dirt sensor, Abstracle sensor

Characteristics of task environment (types)

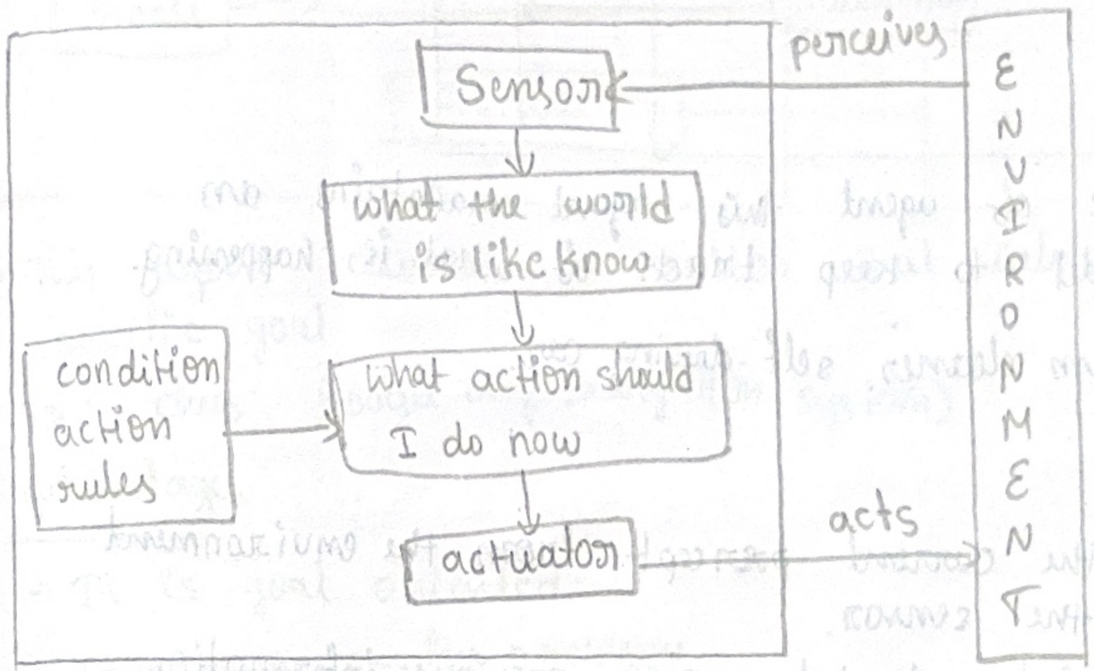
1) fully observable v/s partially observable
2) deterministic v/s stochastic
3) Episodic v/s sequential
4) Static v/s dynamic
5) Discret v/s continuous
6) single agent v/s multiple agent

Environment type	definition	Example
• Fully observable	complete information of the environment is available	chess
• partial observable	some information of the environment is Hidden	self-driving car, Huno
• deterministic	same actions always gives same result	calculator
• stochastic	output is affected by randomness	stock market analysis medical treatment outcomes
• Episodic	Each task is independent	spam email detection
• sequential	current action affects the future	chess
• static	environment does not change	chess board carrom board
• Dynamic	Environment changes continuously	self driving car
• Discret	Finite set of actions	Ticktak
• continuous	Infinite states and action	self driving car
• single agent	only one agent is performing the tasks with no competition	calculator maze solving agent
• multiple agent	Two or more agent working together	Ludo self driving car

Types of AI agent

1) Simple reflex agent
2) model Based agent
3) Goal based agent
4) Utility based agent
5) learning agent

Simple reflex agent



-> It is the type of agent which works only on present situation
-> It works on condition-action rules
Ex:- Automatic doors, thermostat,

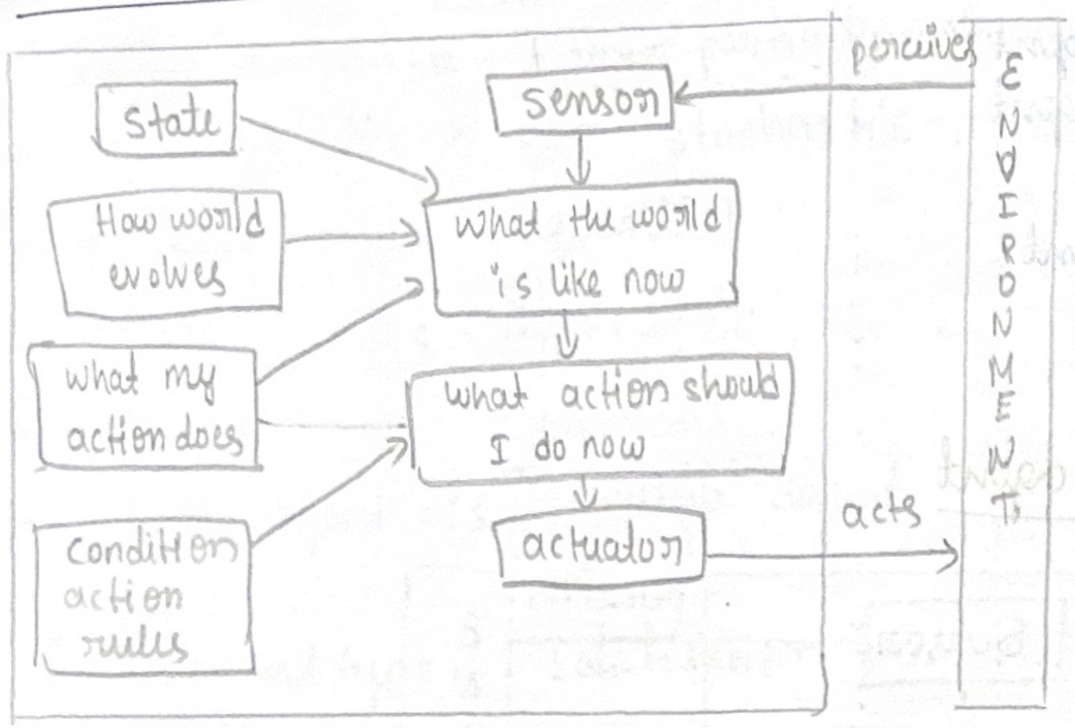
advantages

- It is simple model
- fast decision making.
- It doesn't need memory
- easy to design

disadvantages

- It is doesn't learn from the past experience.
- It doesn't have memory
- It cannot be used to slove the complex problem.

Model based agent



- It is the type of agent this agent maintains an internal model to keep tracks of what is happening.

Ex:- Robot vacuum cleaner, self-driving car.

working

→ It receives the current percept from the environment through the sensor.
→ Updates the internal state using previous information
→ chooses an action using condition action rule.
→ Robot vacuum cleaner remembers ~~nearby vehicles~~ even when the room layout which it already cleaned
→ self driving car remembers nearby vehicles even when temporarily hidden.

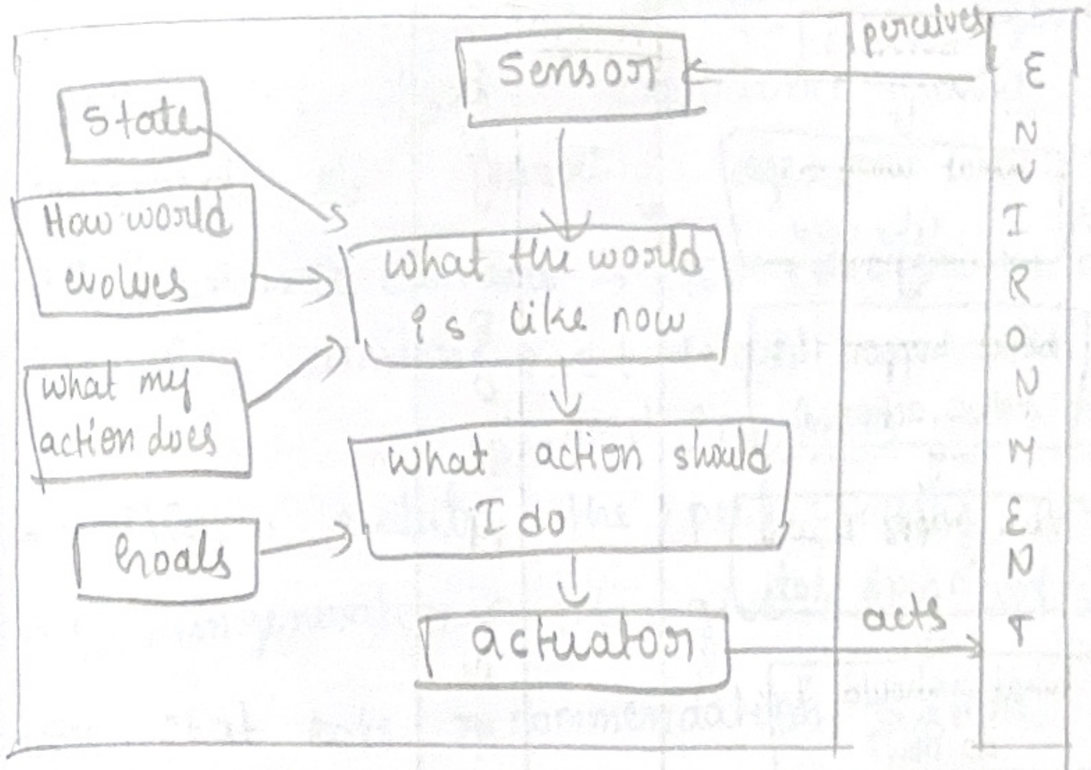
Advantages:-

→ It has memory
→ It is better than simple reflex agent

disadvantages:

- It is more complex to design
- It requires extra memory
- updating the model textime

Goal based agent.



- This agent chooses the action that helps to achieve specific goal.

Ex: chess, Google map (navigation system)

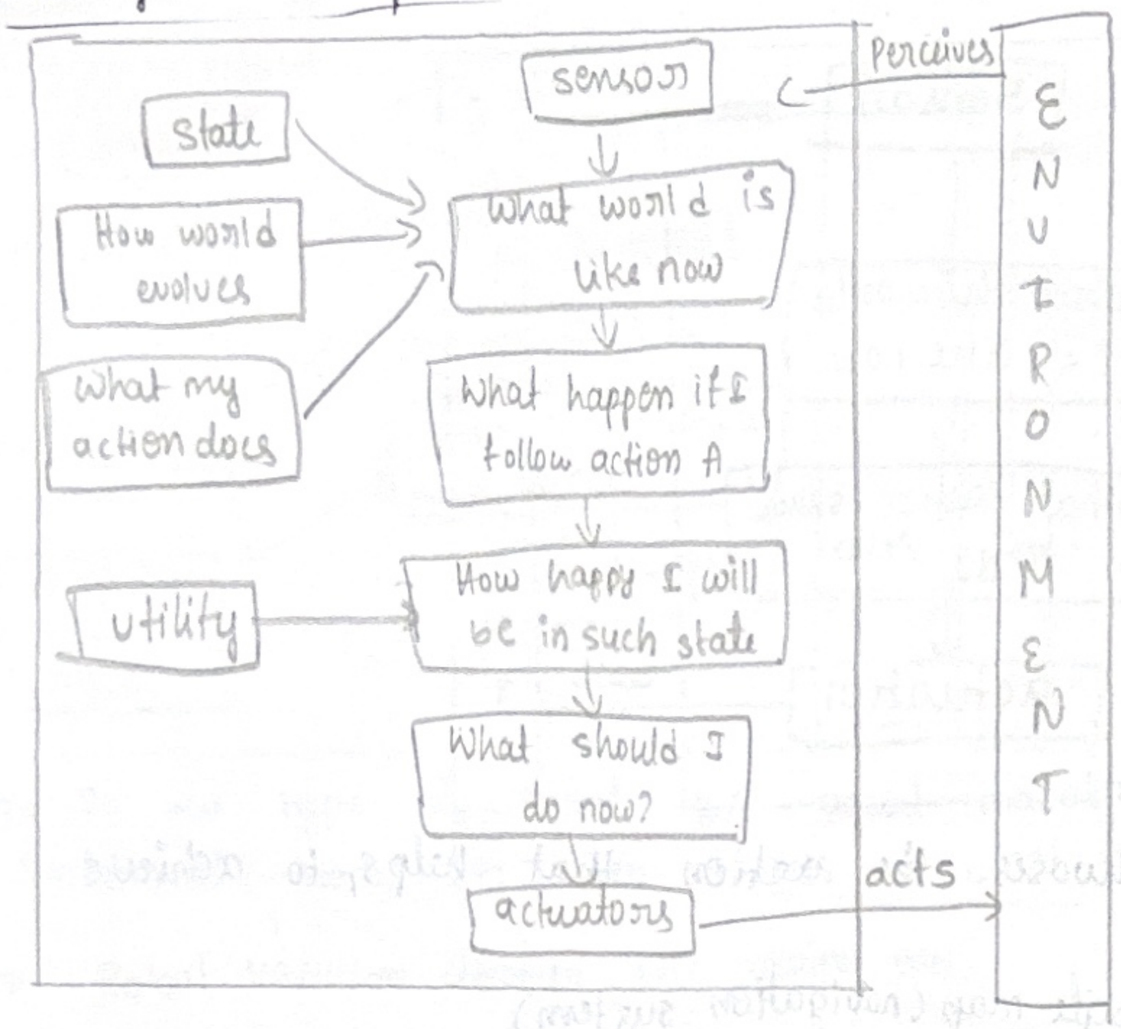
advantages

→ It is goal oriented
→ can solve complex problems

disadvantages

→ It requires planning
→ It can be slower
→ needs more comptational power

utility based agent



utility based agent selects action that provides highest utility when there are multiple ways to active a goal

Utility

It is maximum benifit on the

Ex: self-driving car.

Utility

→ many roots avilable to reaches the destination but self-driving car select fastest, shortest and safest root.

advantages

→ makes the best / optimal decision
→ handles uncertainty
→ considers multiple factors.

disadvantages

→ designing is deficult
→ requires more computation
→ It is complex to implement.

Ex: . self-driving car balancing safety.

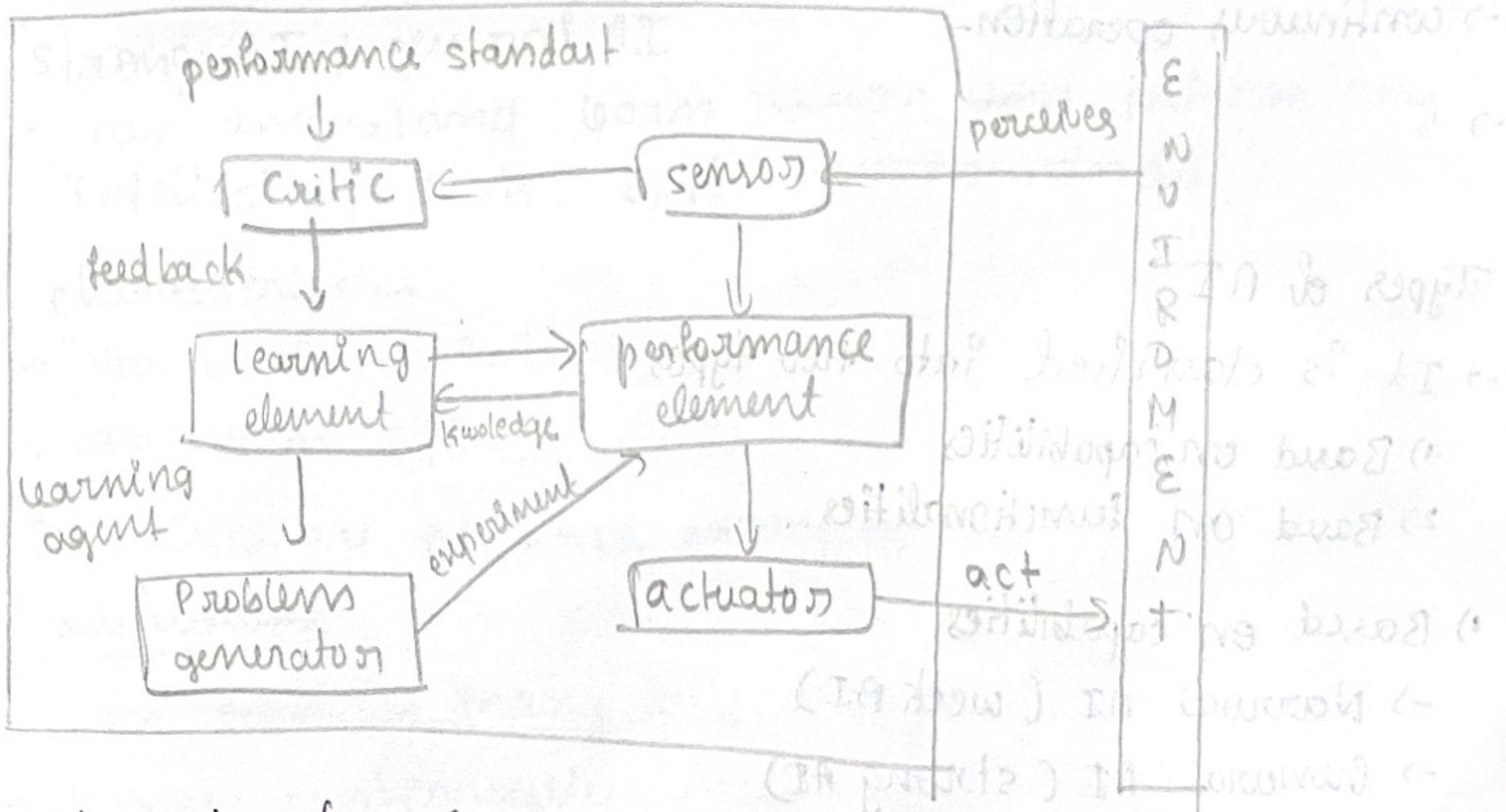
learning base agent

learning agent is an agent which learns from its past experience and feed back.

components of learning agent:

→ Performance element → It selects the actions
→ learning element → It learn from the past experience and improves the performance.
→ critic → evaluates the performance provides feedback
→ Problem generator → It give new action to improve.

Ex: chat bots, recommendation system.



Characteristics of AI.

→ Learning ability → AI learns from data & past experiences
→ Problem solving → AI identifies problems and finds the best possible solutions
→ Decision making → AI evaluates different options and selects the most suitable action.
→ Natural Language processing → AI understands and communicates in human language
→ perception → AI can understand information from images

→ adaptability → AI performs repetitive tasks automatically with little or no human intervention.

→ adaptability → AI adapts to new situations by learning from experience

→ automation → AI performs repetitive tasks automatically with little or no human intervention

→ accuracy → AI performs tasks with high precision and reduces human errors.

→ speed → AI processes large amounts of data quickly and provides fast result.

→ continuous operation-

→

Types of AI

→ It is classified into two types

1) Based on capabilities
2) Based on functionalities.

1) Based on capabilities
   → Narrow AI (week AI)
   → General AI (strong AI)
   → super AI.

Narrow AI (week AI)

→ This is design to perform only one specific task or set of limited task.

→ It cannot perform tasks outside its program area.

characteristics

→ most AI used today are narrow
→ cannot think like human
→ performance only limited set of task

Ex:- ChatGPT, Siri.

advantages

→ fast and efficent
→ highly acurity
→ widle used in todays era

disadvantages

→ No self awareness / human like intellegence
→ cannot perform multiple unrelated task

strong AI / general AI

→ can understand learn reason and perform any intellactual task, that human can do.

characteristics:

→ Human level intellegence
→ can learn different skills.

Ex- There are no true examples existing it

advantages

→ can perform many task
→ learns continuously
→ solves complex problem

disadvantages

→ difficult to develop
→ Ethical and safety concerns
→ High computational requirements.

3. Super AI - It is a theoritical AI that would be more intellegent that human in every field including creativity, reasoning and decision making

Characteristics

→ smarter than human beings
→ self learning and self awareness
ex = not yet existing

advantages

→ Extremly fast and intellegent
→ can be used to solve highly complex global problems

disadvantages

→ difficult to built
→ ethical and security issues

Based on functionalities

1) limited memory
2) reactive mechines
3) Theory of mind
4) self awareness AI

1) limited memory

Uses the Past data along with current information to make decision.

Ex:- Chart GPT, self driving car.

Characteristics

→ It should learn from the Past data.
→ It should also consider current information.

2) reactive machine.

It respose only to the current situtions that have no memory.

Ex:- chess

Characters

→ No learning
→ No memory
→ It uses only current situations

3) Theory of mind

AI can understand human emositions, belives and intensions

characters

→ It should be able to interacte socialy.
→ Understand emositions and mental state

Ex:- stile under reserch.

4) self aware AI

AI with consciousness and self awareness.

Ex:- It is hypothetical that doesnot exists characters

→ It should have emotion & awareness
→ understand its own existence

Approaches of AI

It is a method used to design intellegent system

1) Thinking humanly approach (cognitive modeling)
2) Acting humanly approach (turing test)
3) Thinking rationally approach (laws of thought) (thinking logically) Ex: expert system
4) Acting rationally approach (choosing best performance) self driving car

1) Thinking humanly

This approach aims to make machines that think like humans, by imimitating human thought process.

Characters

-> It is based on human thinking
-> It uses Psycology and cognitive science
-> It focuses on how human slove problem

Ex:- virtual assitance, intellegent tutoring system

advantages

-> It mimix the human thing naturaly
-> It helps to understand the human intellegent

Disadvantage

-> Human thinking is difficult to model acurately
-> It requires extensive knowledge of psycology

Acting humanly.

It approach focuses on making machines believe like humans regardless how they think.

characters

-> It is based on turing test
-> It is mostly based on human like ~~believe~~ behaviour and comunication
-> It requirs NLP, resioning and learning

Ex:- Chart GPT,

advantages

-> It enable naturaly interaction with humans
-> It is useful in coustomer sevice and virtual aissance

Thinking Rationally Approach

Difference between AI, and machine learning and deep learning

Characteritics	AI	ML	DL
Definition	It is the broad field of creating machines that can perform tasks regarding human intelligence.	A subset of AI where systems learn from data instead of being explictly program	deep learning is subset of ML that uses artifical neural networks with many layers
goal	Think and act intelligently	Enables machine to learn patterns from the data	solve complex problems by automatically learning feature from last data
Data requirement	may or may not require data	requires data	requires very large amount of data.
Human involve -ment.	Human involvement is required high (rules can be programmed)	moderate	low (learns features auto matically)
Example	virtual assistance chess & export systems	spam detection, recommendation system & fraud detection	facial regonion self driving car



Note

- All ML is AI
- All DL is ML and AI
- But not all AI is ML or DL

Example

Imagine you want a computer to recognize cats in a image

- AI - (It should) the over all goal is to make computer identity cats.
- ML - you provids thousands of labelled cat and non cat images, and computer learns the pattern, to classify the New images.
- DL - instead of manually telling computer what features to look for a deep neural network automatically learns these features from millions of images.

All that glitters not gold.